- Vietnam has considered establishing nuclear power generation since 1995, and firm proposals surfaced in 2006.
- Russia had agreed to finance and build 2400 MWe of nuclear capacity from 2020, and Japan had agreed similarly for another 2200 MWe.
- In November 2016 plans were cancelled in favour of gas and coal.

Privacy - Terms

- Electricity use is growing at about 10% per year.

Electricity sector

Total generation (in 2023): 274.0 TWh

Generation mix: coal 123 TWh (45%); hydro 80.6 TWh (29%); solar 27.2 TWh (10%); natural gas 26.4 TWh (10%); wind 11.6 TWh (4%); oil 3.5 TWh; biofuels 2.1 TWh.

Import/export balance: 3.5 TWh net import (4.2 TWh imports, 0.7 TWh exports)

Total consumption: 246 TWh

Per capita consumption: c. 2400 kWh in 2023

Source: International Energy Agency, The World Bank. Data for year 2023

Vietnam has a population of about 100 million growing at about 1% per year. Per capita electricity consumption of 2400 kWh in 2023 was over eight times higher than in 2000. Rapid electricity demand growth of up to 10% per year is resulting in rationing. A 500 kV grid runs the length of the country.

Nuclear power proposals

In the early 1980s two preliminary nuclear power studies were undertaken, followed by another which in 1995 reported: "Around the year 2015, when electricity demand reaches more than 100 billion kWh, nuclear power should be introduced for satisfying the continuous growth in the country's electricity demand in that time and beyond."

In February 2006 the government announced that a 2000 MWe nuclear power plant should be on line by 2020. This general target was confirmed in a nuclear power development plan approved by the government in August 2007, with the target being raised to a total of 8000 MWe nuclear by 2025. A general law on nuclear energy was passed in mid-2008, and a comprehensive legal and regulatory framework was then being developed.

Two reactors totalling 2000 MWe were proposed at Phuoc Dinh in the southern Ninh Thuan province, as well as a further twin-unit

2000 MWe plant at Vinh Hai nearby. These two plants were to be followed by a further 6000 MWe by 2030. Both locations were chosen based particularly on geological suitability on the coast. A high demand scenario suggested 8000 MWe in 2025 and 15,000 MWe (10% of total) in 2030 at up to eight sites in five provinces, requiring four more units to be added to the first two sites, then six more at three or four central sites in provinces of Quang Ngai (Duc Thang or Duc Chanh), Binh Dinh (Hoai My) and Phu Yen (Xuan Phuong).

Atomstroyexport, Westinghouse, EdF, Kepco, and China Guangdong Nuclear Power Group (CGNPC) all expressed strong interest in supplying the first two twin-unit plants. Unconfirmed reports early in 2010 said that the Japanese government, with Tepco and others, offered an $11 billion contract, also that a consortium of Mitsubishi, Toshiba and Hitachi bid for the project. The plants were to be state-owned under Electricity of Vietnam (EVN), with no private equity.

Formerly proposed nuclear power reactors

Location	Plant (province)	Type	MWe nominal
Phuoc Dinh	Ninh Thuan 1-1	VVER-1200/V-491	1200
	Ninh Thuan 1-2	VVER-1200/V-491	1200
	Ninh Thuan 1-3	VVER-1200/V-491	1200
	Ninh Thuan 1-4	VVER-1200/V-491	1200
Vinh Hai	Ninh Thuan 2-1	Atmea1 ?	1100
	Ninh Thuan 2-2	Atmea1 ?	1100
	Ninh Thuan 2-3	Atmea1 ?	1100
	Ninh Thuan 2-4	Atmea1 ?	1100
Central		APR-1400?	1350
Central		APR-1400?	1350

In July 2011 the government issued a master plan specifying Ninh Thuan 1&2 nuclear power plants with a total of eight 1000 MWe-class reactors, one coming on line each year 2020-27, then two larger ones to 2029 at a central location. The Ministry of Industry & Trade (MOIT) is responsible for the actual projects, while the Ministry of Science & Technology (MOST) supports the program, developing a master plan and regulation.

In January 2014 the government said that nuclear power development would be delayed up to four years, due to continuing negotiations on technology and financing. EVN earlier reported that the IAEA had urged some delay to allow fuller preparation. In January 2015 the VAEA announced a further delay, giving construction start about 2019. In March 2016 the government said that the first reactor would be commissioned in 2028, in line with revisions to the National Power Development Plan to 2020, which also projected to 2030 when 4600 MWe of nuclear capacity was expected online.

However, in November 2016 the National Assembly passed a resolution to postpone indefinitely the plans for the two nuclear power stations, “due to economic conditions in our country today,” and lower demand projections. They will be replaced with 6 GWe of LNG and coal by 2030, based on short-term cost considerations relating to those fuels. Imports of power especially from Laos and also renewables would supplement this.

Earlier in February 2014 Doosan Heavy Industries Vietnam Ltd (Doosan Vina) received ASME certification to manufacture nuclear components, the first company in Southeast Asia to achieve this. The company was established in 2007.

In March 2022 the Ministry of Industry and Trade issued a draft development plan calling for the inclusion of SMRs in the country's energy mix after 2030.

In June 2022 the country’s Minister of Industry and Trade told the National Assembly that “developing nuclear power is an ongoing inevitable trend in the world.” He noted that whilst the government is committed to boosting renewable energy capacity the country would lack a “stable energy source."

In June 2024 a memorandum was signed between Rosatom and Vietnam relating to the construction of the proposed Centre for Nuclear Science and Technology.

In February 2025, Vietnam’s prime minister directed state-owned companies to accelerate construction of two nuclear power plants in Ninh Thuan province, aiming for completion by 2030.

Ninh Thuan 1: Phuoc Dinh (Russia)

A pre-feasibility study for this plant carried out by the MOIT was approved by the National Assembly in November 2009, and a comprehensive feasibility study followed. In May 2010 the prime

minister established the Ninh Thuan nuclear power project. The Ninh Thuan Project Management Board functioned under EVN.

In October 2010 an intergovernmental agreement was signed for Atomstroyexport to build the Ninh Thuan 1 nuclear power plant, using two VVER-1000 reactors based on those at Tianwan in China. However, in October 2014 it was decided to use the later AES-2006 nuclear plants with VVER-1200 reactors. It was to be constructed as a turnkey project. In July 2015 EVN and NIAEP-Atomstroyexport signed a general framework agreement for construction of the first unit, with the actual VVER-1200/V-491 reactors to be supplied by Atomproekt based in St Petersburg. NIAEP-ASE is the parent company of Atomenergoproekt (AEP) which was involved in the design.

Russia's Ministry of Finance was to finance at least 85% of the plant, and to supply the fuel and take back the used fuel for the life of the plant, as is normal Russian policy for non-nuclear-weapons states. An agreement for up to $9 billion finance was signed in November 2011 with the Russian government's state export credit bureau, and a second agreement for a $500 million loan covered the establishment of the Centre for Nuclear Energy Science & Technology (CNEST) jointly by Rosatom and MOST. It seems that all financing was to be in rubles and dong, not US dollars. In 2019 an MoU was signed between Rosatom and Vietnam's Ministry of Science and Techniology for the implementation of CNEST.

In 2014, 344 Vietnamese undergraduate and graduate students were studying in Russia, to prepare for the project, and 150 engineers were helping with the construction of Rostov nuclear plant in Russia. Early in 2017, 28 Vietnamese became the first international graduates of a six-year course on nuclear technology at Russia's National Research Nuclear University MEPhI. For international ventures, Rosatom arranges both university and in-company training for nationals.

The initial schedule was for construction start in 2014 and operation from 2020, but commencement was delayed to 2019, with six years' construction envisaged. Considerable work was done at the site before the project's indefinite postponement, including relocating two villages with much upgrading of facilities and infrastructure.

Earlier in 2012 Rosatom helped establish a Nuclear Industry Information Centre (NIIC) at the Hanoi University of Science & Technology (HUST) based on others in 17 Russian regions. The Hanoi NIIC is the first of six international centres (Astana, Dhaka,

Hanoi, Mersin, Minsk, Istanbul). It catered for 1600 visitors per month, 1300 of these being school students, and 90% being from outside Hanoi.

Ninh Thuan 2: Vinh Hai (Japan)

In October 2010 an intergovernmental agreement with Japan was signed for construction of a second nuclear power plant at Vinh Hai in Ninh Thuan province, with its two reactors to come on line in 2024-25. The following month the government signed a further accord with Japan on this, and Japan's Ministry of Economy, Trade and Industry (METI), said that Japan Atomic Power Co. (JAPC) and the International Nuclear Energy Development of Japan Co. Ltd. (JINED), would work with EVN on the project, which was to involve financing and insurance of up to 85% of the total cost. JINED is a consortium of Japan's METI, nine utilities (led by Chubu, Kansai & Tepco) and three manufacturers (Mitsubishi Heavy Industries, Toshiba and Hitachi). The intergovernmental agreement took effect from January 2012.

In February 2011 JAPC signed an agreement with EVN to advance the feasibility study, and in September it signed a contract with EVN to provide consulting services to help with site selection and an 18-month, $26 million feasibility study including technology selection with economic and financial analysis, funded by the Japanese government. The following day an MoU was signed between EVN and JINED to progress the design, construction and operation of the plant. EVN listed six criteria to apply, including late-model reactors, stable supply of fuel, support for local industry and education of staff, and financial support. Japan committed to train about 1000 staff for Ninh Thuan 2, and in January 2015 Westinghouse committed to train staff too.

Vinh Hai is on Cam Ranh Bay, about 20 km northeast of Phuoc Dinh. In October 2011 a less-formal arrangement for cooperation in construction was signed, and work on the infrastructure for construction was under way. JAPC's feasibility study for EVN was completed in May 2013. A decision on technology was not made, though PWR appeared likely, and Mitsubishi earlier said that if it is PWR then the reference plant would be Hokkaido's Tomari 3, an 866 MWe unit. Toshiba through Westinghouse was keen to supply AP1000 units. However, in September 2015 industry sources said Atmea1 was likely to be chosen.

South Korea and other links

The Vietnam and South Korean presidents in November 2011 approved a joint plan on nuclear power plant construction which they agreed to use as a basis for future cooperative projects. In March 2012 a nuclear cooperation agreement was signed to take this forward, with a one-year feasibility study on constructing a Korean plant. In June 2013 a joint preliminary feasibility study commenced "for nuclear plant development worth $10 billion", and both governments agreed to cooperate further on the development project.

A nuclear cooperation agreement was signed with Russia in 2002, and since 2006, others were signed with France, China (in particular with CGNPC), South Korea, Japan, USA and Canada. In 2007 there was an agreement between the US Department of Energy's (DoE's) National Nuclear Security Administration (NNSA) and Vietnam's MOST for cooperation and information exchange on the peaceful uses of nuclear energy. A further nuclear agreement with the USA was signed in March 2010, then a full cooperation and commercial trade agreement was initialed in October 2013 and approved by Congress in May 2014 allowing US companies to do business with Vietnam. Earlier in June 2010 the Japan Atomic Energy Agency signed an agreement with the Vietnam Agency for Radiation and Nuclear Safety & Control (VARANS) for infrastructure development for safeguards and nuclear security in respect of nuclear nonproliferation.

Uranium and fuel cycle

Since 2010, the Ministry of Natural Resources & Environment's Department of Geology & Minerals has undertaken evaluation of a uranium deposit in Quang Nam province which is believed to have about 7000 tU in 0.05% ore. Canadian company NWT Uranium Corp was asked in 2010 to help assess prospects.

Vietnam's plans called for importing all the fuel required for reactors. The 2010 agreement with the USA expressed Vietnam's intent to rely on international markets for nuclear fuel supplies and not to pursue domestic enrichment capabilities.

Research and development

Rosatom was working with MOST to establish the Centre for Nuclear Energy Science & Technology (CNEST) under VINATOM, based in Hanoi. Agreements were signed on this in June 2017 and in 2019. A multi-purose 15 MWt VVR research reactor was planned, as well as a complex of laboratories and engineering infrastructure. Rosatom has said that deferral of plans for nuclear power will not affect its R&D commitment. However, the research reactor was to be used for training staff for the nuclear power programme, and so it is unlikely to now proceed.

An earlier nuclear cooperation agreement with Russia relates principally to Vietnam's 500 kW Da Lat research reactor in Lam Dong province, built in 1980, commissioned 1984. This replaced an earlier US Triga MkII reactor which started in 1963 but was dismantled by the USA in the early 1970s. In 2007 the USA helped convert the Da Lat reactor to use low-enriched fuel. It is run by VINATOM, but is due to be shut down about 2025. It is 300 km north of Ho Chi Minh city.

The Dong Nai Research Centre is in the south at Ho Chi Minh city (Saigon), and was to assume more importance under CNEST. The Institute of Nuclear Technology is in the north.

In late 2014 Hanoi University of Science and Technology and Electric Power University signed agreements with GE Hitachi Nuclear Energy (GEH) to cooperate in nuclear engineering. Early in 2015, GEH also signed an agreement with VARANS to promote human resources training and development in the field of nuclear safety analysis for boiling water reactors. In October 2015 a further agreement was signed between GEH and VAEA to enhance its understanding of light water reactor technology and nuclear project management and train staff.

Organization and regulation

Vietnam's new Atomic Energy Law was passed in June 2008 and came into effect early in 2009. Under this, a national nuclear safety commission responsible to the Prime Minister for safety and licensing was established in July 2010.

The Vietnam Atomic Energy Commission/ Agency (VAEA) was established in 1976 and is under the Ministry of Science & Technology (MOST). A national steering committee with the role of Implementing Organisation, and including the representatives of the different ministries and governmental organizations, was established in May 2010 by the Prime Minister.

The Vietnam Atomic Energy Institute (VINATOM) is part of MOST, and responsible for R&D, technical support, personnel training and technical services including the Da Lat research reactor.

The Vietnam Agency for Radiation and Nuclear Safety & Control (VARANS) is the regulator, also part of MOST. In June 2008 the US Nuclear Regulatory Commission and VARANS signed a cooperation agreement to share technical information on nuclear energy as well as exchanging information about regulations, environmental impact and safety of nuclear sites. This was the primary formative influence on VARANS, though it has also actively cooperated with NISA (Japan) and Rostechnadzor (Russia). In February 2015 VARANS signed an agreement with GE Hitachi Nuclear Energy (GEH), for it to train and develop people in the field of nuclear safety analysis for boiling water reactors.

In May 2013 the prime minister announced the establishment of a new National Council for Atomic Energy Development & Application which was to identify strategies and priorities and advise the government. It was to coordinate ministries, agencies, governmental bodies and localities in developing nuclear energy and "realizing" nuclear power programs. It was also to take on a role in international nuclear cooperation activities with organizations and individual countries.

Non-Proliferation

Vietnam's safeguards agreement with the IAEA under the NPT entered force in 1990. and the Additional Protocol has been in force since 2012.

Notes & references

General sources

Le, Doan Phac, 2011, Vietnam's Nuclear Power Development Plan: Challenges and Preparation Work for the First Nuclear Power Projects, INPRO SMR Forum Oct 2011
Council for atomic energy development set up, Socialist Republic of Viet Nam Government News (9 May 2013)
Nagibina, E, *Workforce Planning for Nuclear Power Programs of Rosatom Partner Countries*, Forum of Nuclear Industry Suppliers

ATOMEX Asia 2014 held in Ho Chi Minh City, Vietnam in November 2014

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